



Amazon DocumentDB

Lecture Slides

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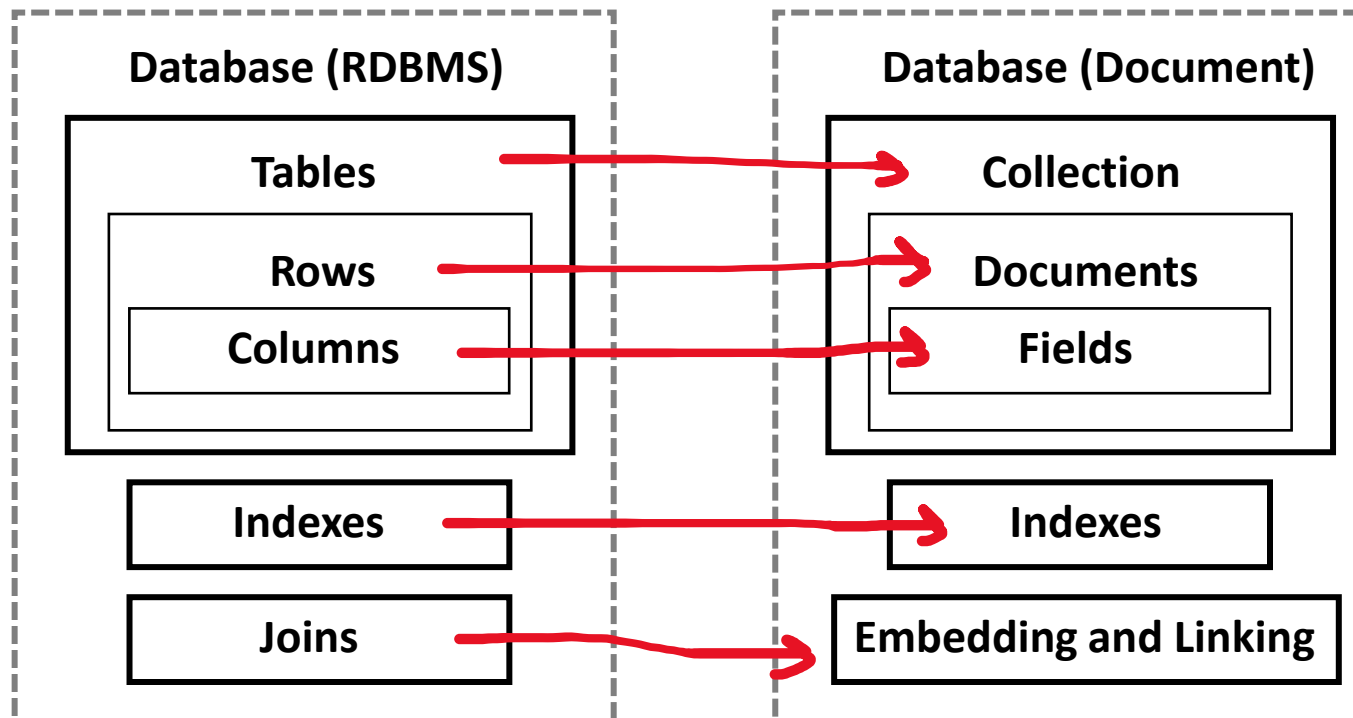
What is a Document store?

A **document store** is a NOSQL database that stores **documents** as its primary data structure.

A document could be an XML but more commonly is JSON or JSON-Like

Document stores are sub-class of Key/Value stores

The components of a document store compared to Relational database



MongoDB



MongoDB is an open-source document database **which stores JSON-like documents**
The primary data structure for MongoDB is called **BSON**

Binary JSON (BSON)

- BSON is a binary representation of JSON-like documents
- BSON is designed to be **efficient both in storage space and scan-speed** compared to JSON
- BSON has more data-types than JSON:
 - Eg. Datetime, byte arrays, regular expressions, MD5 binary data, javascript code

```
BSON:
\x16\x00\x00\x00      // total document size
\x02                   // 0x02 = type String
hello\x00              // field name
\x06\x00\x00\x00world\x00 // field value (size of value, value, null terminator)
\x00                   // 0x00 = type E00 ('end of object')
```

What it looks like to perform an operation on a MongoDB databases

```
db.inventory.insertMany([
  { item: "journal", qty: 25, size: { h: 14, w: 21, uom: "cm" }, status: "A" },
  { item: "notebook", qty: 50, size: { h: 8.5, w: 11, uom: "in" }, status: "A" },
  { item: "paper", qty: 100, size: { h: 8.5, w: 11, uom: "in" }, status: "D" },
  { item: "planner", qty: 75, size: { h: 22.85, w: 30, uom: "cm" }, status: "D" },
  { item: "postcard", qty: 45, size: { h: 10, w: 15.25, uom: "cm" }, status: "A" }
]);
```

MongoDB

- MongoDB uses with a interactive shell (mongosh) or a MongoDB driver to interact with MongoDB
 - MongoDB traditionally doesn't use an SQL language but there is MQL and Atlas SQL
- Default Port for MongoDB: **27017**
- MongoDB supports searches against:
 - fields
 - ranged queries
 - regular-expressions
- MongoDB supports **primary** and **secondary** indexes
- High availability can be obtained via replica sets (replica to offload reads or acts a stand-by in case of failover)
- MongoDB scales horizontally using sharding
- MongoDB can run over multiple servers via load balancing
- MongoDB can be used as a file system, called **GridFS**
 - with load balancing and data replication features over multiple machines for storing files.
- MongoDB provides three ways to perform aggregation (grouping data during a query)
 - aggregation pipeline
 - map-reduce
 - single-purpose aggregation
- MongoDB supports fixed-size collections called capped collections
- MongoDB claims to support multi-document ACID transactions

Amazon DocumentDB



DocumentDB is a NoSQL **document** database that is “MongoDB compatible”

MongoDB is very popular NoSQL among developers. There were open-source licensing issues around using open-source MongoDB, so AWS got around it by just building their own MongoDB database.

When you want a MongoDB database.

Cluster types:

- **Instance based Cluster** — manage your instances directly choosing instance type
 - **Elastic Cluster** — clusters automatically scale, you choose vCPU and number of instances per shard
-
- Compatible with MongoDB 4.0 and 5.0
 - DocumentDB does not support all functionality for MongoDB eg. **Writable Retries is not support**
 - DocumentDB storage volume grows in increments of 10 GB, up to a maximum of 128 TiB.
 - Create up-to 15 replicas
 - Amazon DocumentDB continuously monitors the health of your cluster and automatically restart failed instances
 - Failover automatically will occur to upto 15 replicas in other AZs
 - Backup is turned on by default (can't be turned off) with a retention period between 1 and 35 days
 - Supports point-in-time recovery
 - Clusters are deployed into a customer's VPC
 - Performance Insights feature to determine bottlenecks for reads and writes
 - In-transit and at-rest encryption. You must connect using TLS connect.